Apuntes Parcial Ciencia de Mater...

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Ciencia de Materiales

1º Grado en Ingeniería Aeroespacial

Escuela Técnica Superior de Ingeniería del Diseño
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MATERIALS SCIENCE

Complete Exam Study Guide

Units 2 – 5

Structure of Materials • Crystalline Defects • Mechanical Properties • Plastic Deformation & Strengthening

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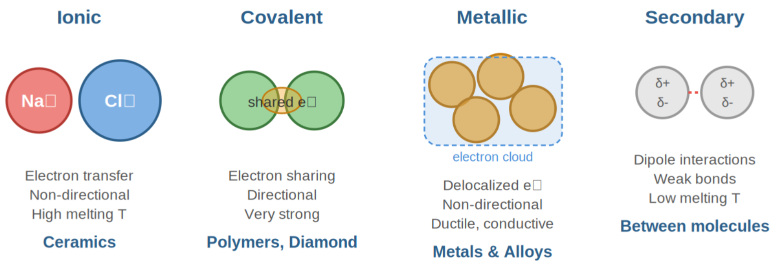
Table of Contents

UNIT 2: STRUCTURE OF MATERIALS

2.1 Atomic Bonding

The properties of engineering materials are determined fundamentally by the type of atomic bonding between their constituent atoms. The Periodic Table helps identify trends in atomic size, melting point, and reactivity. There are four important bonding mechanisms:

Types of Atomic Bonds



2.1.1 Primary Bonds

Ionic Bonds: Found in compounds of metallic + nonmetallic elements (e.g., NaCl, MgO, Al_2O_3). Metallic atoms give up valence electrons to nonmetallic atoms, forming cations and anions. Coulombic attractive forces hold them together. These bonds are nondirectional, have high bonding energies (reflected in high melting temperatures), and produce materials that are hard, brittle, and electrically/thermally insulative.

Covalent Bonds: Found between nonmetallic elements with similar electronegativity (e.g., H_2O, CH_4, diamond, SiC). Valence electrons are shared between atoms to form molecules. These bonds are directional (specific angles, e.g., 109.5° in diamond). Can form gases, liquids, or very strong solids (diamond, silicon carbide). Long molecular chains form polymers.

Metallic Bonds: Found between electropositive (metallic) atoms. Valence electrons are not bound to any particular atom — they belong to the metal as a whole, forming an “electron cloud” that surrounds positive ion cores (cations). Nondirectional bond. Produces ductile solids that are good electrical and thermal conductors. Examples: all metals and their alloys.

2.1.2 Secondary Bonds (Van der Waals)

Weak physical bonds between molecules without electron transfer or sharing. They arise from electric dipoles:

Induced dipole (London forces): Instantaneous asymmetry in electron cloud induces a dipole in a neighboring atom. Found in noble gases (He, Ne, Ar), H_2, Cl_2.

Permanent dipole: Molecules with atoms having large differences in electronegativity (e.g., HCl). The positive end of one molecule attracts the negative end of another.

Hydrogen bond: A special case of permanent dipole. Occurs between molecules where H is covalently bonded to N, O, or F. Very important in water (H_2O), HF, and polymeric chains.

2.1.3 Binding Energy and Interatomic Spacing

Atoms are separated by an equilibrium distance where repulsive and attractive forces balance. At this distance, the interatomic energy (IAE) is at a minimum. The binding energy is the energy required to create or break the bond. Higher binding energy means higher strength and higher melting point.

Bond Type	Example	Energy (eV)	Melting Pt (°C)	Features
Ionic	NaCl / MgO	3.3 / 5.2	801 / 2800	Hard, brittle, insulating

Covalent	Si / C(diamond)	4.7 / 7.4	1410 / >3550	Very strong, directional
Metallic	Al / Fe / W	3.4 / 4.2 / 8.8	660 / 1538 / 3410	Ductile, conductive
Van der Waals	Ar / Cl_2	0.08 / 0.32	-189 / -101	Weak
Hydrogen	H_2O	0.52	0	Important in polymers

2.2 Atomic Order: Crystal Structure

Most engineering materials are crystalline solids. Their properties depend on the crystal structure — the three-dimensional repeating arrangement of atoms. The simplest structural unit that defines the crystal structure is called the unit cell.

2.2.1 Unit Cell Geometry and Lattice Parameters

Most crystal structures are parallelepipeds or prisms with three sets of parallel faces. A unit cell is fully described by six lattice parameters: three edge lengths (a, b, c) and three interaxial angles (α, β, γ).

There are 7 crystal systems (Cubic, Tetragonal, Orthorhombic, Rhombohedral, Hexagonal, Monoclinic, Triclinic) and 14 Bravais lattices.

2.2.2 Counting Atoms in a Unit Cell

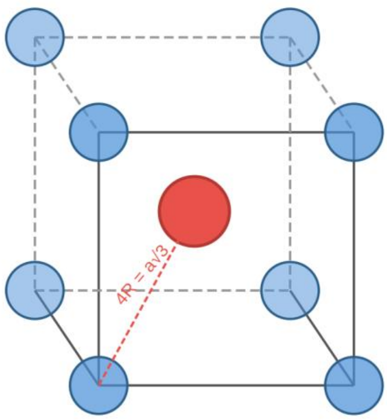
Atoms at different positions contribute different fractions to the unit cell:

- **Corner atom:** contributes 1/8 (shared by 8 unit cells)
- **Face-center atom:** contributes 1/2 (shared by 2 unit cells)
- **Edge atom:** contributes 1/4 (shared by 4 unit cells)
- **Body-center atom:** contributes 1 (entirely inside one cell)
- **Mid-plane atom (HCP):** contributes 1 if fully inside; top/bottom face atoms contribute 1/2 each, corner atoms 1/6 each

2.2.3 Body-Centered Cubic (BCC)

Examples: Cr, V, α-Fe, Mo, W

BCC Unit Cell



Atoms/cell = 2 | CN = 8 | APF = 0.68
a = 4R/√3 (e.g. Fe, Cr, W, Mo, V)

Lattice parameter – atomic radius relation: The body diagonal passes through the center atom. 4R = a√3, so a = 4R/√3

Atoms per unit cell: 1 (center) + 8 × 1/8 (corners) = 2 atoms

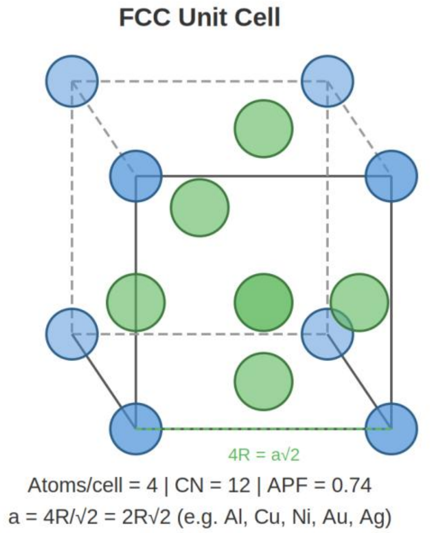
Coordination number: 8 (each atom touches 8 neighbors)

Atomic Packing Factor (APF): 0.68

$$APF = V\_atoms / V\_cell = [2 \times (4/3)\pi R^3] / (4R/\sqrt{3})^3 = 0.68$$

2.2.4 Face-Centered Cubic (FCC)

Examples: γ-Fe, Al, Ni, Cu, Ag, Pt, Au



Lattice parameter – atomic radius relation: The face diagonal is the close-packed direction. 4R = a√2, so a = 2R√2

Atoms per unit cell: 6 × 1/2 (faces) + 8 × 1/8 (corners) = 4 atoms

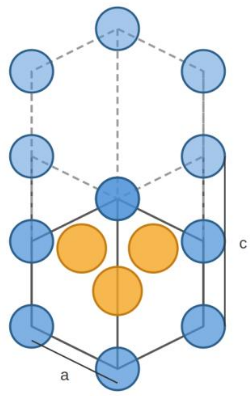
Coordination number: 12

APF: 0.74 (maximum for cubic structures)

2.2.5 Hexagonal Close-Packed (HCP)

Examples: Mg, Zn, Zr, α-Ti, Be

HCP Unit Cell



Atoms/cell = 6 | CN = 12 | APF = 0.74
a = 2R, c/a = 1.633 (e.g. Mg, Zn, Ti, Zr)

Lattice parameter: a = 2R, ideal c/a ratio = 1.633

Atoms per unit cell: 3 (mid-plane) + 2 × 1/2 (top/bottom face centers) + 12 × 1/6 (corners) = 6 atoms

Coordination number: 12

APF: 0.74 (same as FCC — both are close-packed structures)

$$V_{cell} = (3a^2c\sqrt{3}) / 2, \text{ where } c = 1.633a \text{ for ideal HCP}$$

2.2.6 Summary: Crystal Structure Comparison

Property	BCC	FCC	HCP	Simple Cubic
Atoms/cell	2	4	6	1
CN	8	12	12	6
APF	0.68	0.74	0.74	0.52
a vs R	4R/√3	2R√2	a = 2R	a = 2R
Examples	Fe, Cr, W, Mo, V	Al, Cu, Ni, Au, Ag	Mg, Zn, Ti, Zr	Po

2.2.7 Theoretical Density

The theoretical density of a crystalline material is calculated from the unit cell:

$$\rho = (n \times A) / (V_c \times N_A)$$

Where: n = number of atoms per unit cell, A = molar atomic mass (g/mol), V_c = volume of unit cell (cm^3), N_A = Avogadro's number (6.022×10^{23} atoms/mol).

Linear atomic density: number of atoms per unit length whose centers lie on a specific crystallographic direction.

Planar atomic density: number of atoms per unit area centered on a particular crystallographic plane.

2.2.8 Interstitial Sites (Voids)

Interstitial sites are small voids created between atoms when planes stack. Impurities or alloying elements can reside in these sites. The size of the interstitial location is much smaller than the lattice atoms.

Tetrahedral void: formed between 4 atoms (e.g., ZnS structure, Zn^{2+} in tetrahedral sites)

Octahedral void: formed between 6 atoms (e.g., NaCl structure, Na^+ in octahedral sites)

2.3 Crystallographic Directions and Planes (Miller Indices)

2.3.1 Crystallographic Directions [uvw]

A direction is represented by a vector. Steps to determine the Miller indices:

1. Place the origin at a corner of the unit cell.

2. Translate the vector so it passes through the origin.

3. Determine projections on x, y, z axes in terms of a, b, c.

4. Multiply or divide to get smallest integers.

5. Negative indices are shown with a bar over the number.

6. Enclose in square brackets: [uvw].

Example: vector from origin to (1/2, 0, 1) → [1 0 2] (multiply by 2).

2.3.2 Crystallographic Planes (hkl)

Steps to determine plane Miller indices:

1. If the plane passes through the origin, translate it or choose a new origin.

2. Find where the plane intercepts each axis (in terms of a, b, c).

3. Take reciprocals of the intercepts. A plane parallel to an axis has infinite intercept → index 0.

4. Clear fractions to get smallest integers.

5. Enclose in parentheses: (hkl).

Example: plane intercepts x=1, y=2, z=∞ → reciprocals: 1, 1/2, 0 → clear fractions: (2 1 0).

2.4 Structure of Ceramics

2.4.1 Crystalline Ceramics

Ceramics are made of 2+ elements (higher complexity than metals). Bonding is ionic: metal → cation (+), nonmetal → anion (-). Ionic crystal structures can be viewed as close-packed anion structures, with cations fitting into interstitial sites.

Key factors: 1) Ionic Radii — the Rc/Ra ratio determines coordination geometry; 2) Electrical neutrality must be maintained; 3) Anion polyhedra shape depends on Rc/Ra.

CN	Rc/Ra Range	Geometry
2	< 0.155	Linear
3	0.155 – 0.225	Triangular
4	0.225 – 0.414	Tetrahedral
6	0.414 – 0.732	Octahedral
8	0.732 – 1.0	Cubic

2.4.2 Common Ceramic Structures

AX-type (1:1 ratio): CsCl (Rc/Ra=0.939, CN=8); NaCl (Rc/Ra=0.564, CN=6, also MgO, FeO, CaO, TiC); ZnS blende (Rc/Ra=0.402, CN=4, also SiC, ZnO).

AmXp-type (different ratio): Fluorite CaF_2 (Ca^{2+} at FCC positions, F^- at ALL tetrahedral sites, CN of F=4, CN of Ca=8). Also CeO_2, ZrO_2.

AmBnXp-type (multiple cations): Perovskite structure ($CaTiO_3$, $BaTiO_3$, $PbZrO_3$). Ba^{2+} at corners, O^{2-} at face centers, Ti^{4+} at body center. Below 120°C, $BaTiO_3$ shows spontaneous polarization → piezoelectric and ferroelectric properties.

Piezoelectricity: Materials that develop voltage upon application of stress (direct effect) and develop strain when an electric field is applied (converse effect). Key for sensors and actuators.

2.4.3 Silicate Structures

Silicates are composed of silicon and oxygen (the two most abundant elements on Earth). The basic unit is the SiO_4^{4-} tetrahedron with strong covalent Si-O bonds.

Chain silicates: Tetrahedra linked in chains (e.g., $MgSiO_3$ soapstone)

Ring silicates: Tetrahedra in rings (e.g., $Be_3Al_2(Si_6O_{18})$ emerald)

Laminar (sheet) silicates: 2D layers (e.g., kaolinite)

Spatial (3D) silicates: Quartz SiO_2 (crystalline) or fused silica (amorphous/glass). Addition of Na_2O/CaO creates glass.

2.4.4 Other Crystalline Materials: Diamond and Graphite

Diamond: Strong covalent C–C bonds (4 bonds per atom, tetrahedral 109.5°), one of the highest melting points (3550°C), extremely hard, electrical insulator, high thermal conductivity.

Graphite: Strong covalent bonds within layers (3 bonds per atom, hexagonal pattern), weak van der Waals between layers. Electrically conductive (4th electron free). Lower density than diamond.

2.4.5 Allotropy (Polymorphism)

Some materials exist in different crystal structures depending on temperature or pressure:

Carbon: graphite / diamond / graphene / fullerenes / nanotubes

Iron: BCC (α-Fe, room T) → FCC (γ-Fe, 912°C) → BCC (δ-Fe, 1394°C)

Titanium: HCP (α) → BCC (β)

2.5 Structure of Polymers

Polymer: Long carbon chains (macromolecules) formed from small molecules called monomers. "Poly" = many, "mer" = unit.

Molecular Weight: Properties depend heavily on chain length. Low MW → gases/liquids; medium MW → waxes; high MW → solid polymers.

Degree of polymerization: Average molecular weight of polymer divided by the molecular weight of the monomer (number of repeating units).

Crystallinity: Observed in zones where chains are ordered. Polymers can be amorphous (disordered), semi-crystalline (ordered + disordered regions). Key thermal transitions: glass transition temperature (Tg, amorphous) and melting temperature (Tm, crystalline).

2.6 X-Ray Diffraction (XRD)

Bragg's Law: Constructive interference occurs when:

$$n\lambda = 2d\,\sin\theta$$

Where: n = order (usually 1), λ = X-ray wavelength, d = interplanar spacing, θ = diffraction angle.

Interplanar spacing (cubic):

$$d(hkl) = a \,/\, \sqrt{(h^2 + k^2 + l^2)}$$

2.6.1 Diffraction Rules for Cubic Lattices

BCC: h + k + l must be EVEN (= 2n). Diffracting planes: (110), (200), (211), (220), (310)...

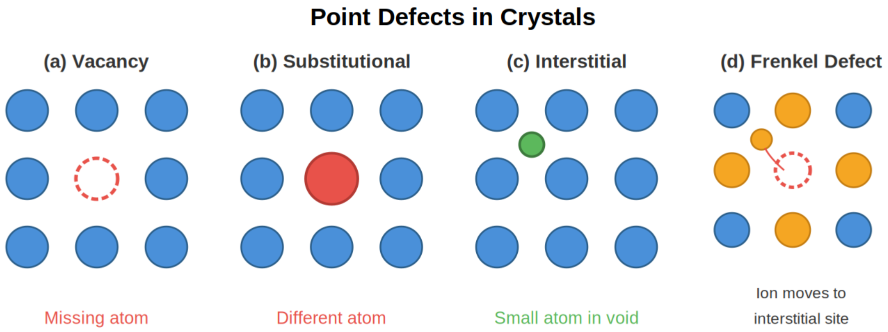
FCC: h, k, l must be ALL ODD or ALL EVEN (0 is even). Diffracting planes: (111), (200), (220), (311)...

Distinguishing BCC from FCC: Compare d-spacing ratio of first two peaks. For BCC: d_2/d_1 = d(200)/d(110) = 0.707. For FCC: d_2/d_1 = d(200)/d(111) = 0.866.

UNIT 3: CRYSTALLINE DEFECTS

3.1 Point Defects

Point defects are imperfections located at one (or a few) sites in the crystal. They disrupt the perfect arrangement of surrounding atoms.



3.1.1 Vacancy

A vacant atomic site in the crystal lattice. Produced by temperature increases, plastic deformation, solidification, or high-energy radiation.

Vacancy concentration (Arrhenius equation):

$$N_v / N = exp(-E_v / k_B T) \quad or \quad N_v / N = exp(-Q_v / RT)$$

Where N_v = number of vacancies/cm³, N = number of atoms/cm³, E_v = energy per vacancy, k_B = Boltzmann constant, Q_v = activation energy per mole, R = gas constant (8.314 J/mol·K), T = temperature (K).

Vacancy concentration increases exponentially with temperature.

Ionic vacancies: Frenkel defect (cation-vacancy + cation in interstitial site); Schottky defect (paired cation + anion vacancies).

3.1.2 Solute Defects

Foreign atoms dispersed in the lattice forming a solid solution (original lattice is maintained):

Substitutional: A foreign atom replaces a lattice atom at its site. Can have unlimited or limited solubility depending on Hume-Rothery rules (atom size, crystal structure, valence, electronegativity). E.g., Cu-Ni.

Interstitial: A much smaller atom occupies an interstitial site. Always limited solubility. E.g., carbon in iron (Fe-C).

3.2 Diffusion

Atomic movement through matter. Driven by concentration differences. Rate: Gases > Liquids > Solids. Increases with temperature.

3.2.1 Fick's First Law (Steady-State Diffusion)

$$J = -D \times (dC/dx)$$

Where: J = flux (atoms/cm²·s), D = diffusion coefficient (cm²/s), dC/dx = concentration gradient.

Diffusion coefficient:

$$D = D_0 \times exp(-Q_d / RT)$$

Where D_0 = pre-exponential constant, Q_d = activation energy for diffusion, R = 8.314 J/mol·K, T = temperature (K). Plotting ln(D) vs 1/T gives a straight line: slope = -Q_d/R, intercept = ln(D_0).

3.2.2 Fick's Second Law (Non-Steady-State Diffusion)

$$\partial C/\partial t = D \times (\partial^2 C/\partial x^2)$$

For semi-infinite solid with constant surface concentration (e.g., carburizing):

$$(C\_x - C_0) / (C\_s - C_0) = 1 - erf(x / (2\sqrt{(Dt)}))$$

Where: C_x = concentration at depth x after time t, C_0 = initial uniform concentration, C_s = constant surface concentration, erf = Gaussian error function (tabulated).

Carburizing: Surface treatment of steel to increase surface hardness. Steel placed in carbon-rich atmosphere at high temperature. Carbon diffuses inward. The erf function and diffusion data allow calculation of time/temperature needed for a desired carbon profile.

Key exam skill: Given C_0, C_s, C_x, x, and T → find t (or vice versa). Steps: 1) calculate the erf argument; 2) use the erf table with interpolation to find z; 3) solve z = x/(2√(Dt)) for the unknown.

3.3 Line Defects: Dislocations

3.3.1 Types of Dislocations

Edge dislocation: An extra half-plane of atoms inserted into the crystal. The bottom edge of this extra plane is the dislocation line. Represented by the symbol ⊥. Burger's vector b is perpendicular to the dislocation line.

Screw dislocation: Formed by shearing one part of the crystal relative to another along a line. Burger's vector is parallel to the dislocation line.

Mixed dislocation: Has both edge and screw components with a transition region.

3.3.2 Dislocations and Plastic Deformation

Plastic deformation in metals occurs by the coordinated movement of a great number of dislocations. This explains why the actual shear stress needed for plastic deformation is orders of magnitude lower than the theoretical shear stress (τ_theoretical = G×b/(2πa) ≫ τ_experimental).

Slip system = slip plane + slip direction. Dislocations move on the densest planes along the densest directions.

Structure	Slip Planes	# Slip Systems	Ductility	Cross-slip?
FCC	{111}	12	Very ductile	Can occur
BCC	{110},{112},{123}	12 (48 total)	Ductile, stronger	Can occur
HCP	{0001} basal	3	Low ductility	Cannot

3.4 Surface/Interfacial Defects

3.4.1 Grain Boundaries

Most crystalline solids are polycrystalline (multiple grains in different orientations). Grain boundaries are the interfaces between grains where atoms are disordered. Single crystals are anisotropic; polycrystalline materials are isotropic (on average).

3.4.2 Twinning

An alternative deformation mechanism. Application of stress displaces atoms to form a mirror image (twin) across a boundary plane. The crystal deforms as a result.

3.4.3 Grain Size Measurement (ASTM)

$$n = 2^{(N-1)}$$

Where n = number of grains per square inch at 100× magnification, N = ASTM grain size number.

3.4.4 Microscopy Techniques

Optical microscopy: Reflected light, reveals grain boundaries, grain size, shape (dendritic, equiaxial, acicular), and phases (monophasic, biphasic, multiphasic).

TEM (Transmission Electron Microscopy): Observes dislocations, grain boundaries at atomic level.

SEM (Scanning Electron Microscopy): Fracture surfaces, 3D topography.

UNIT 4: MECHANICAL PROPERTIES

4.1 Stress and Strain

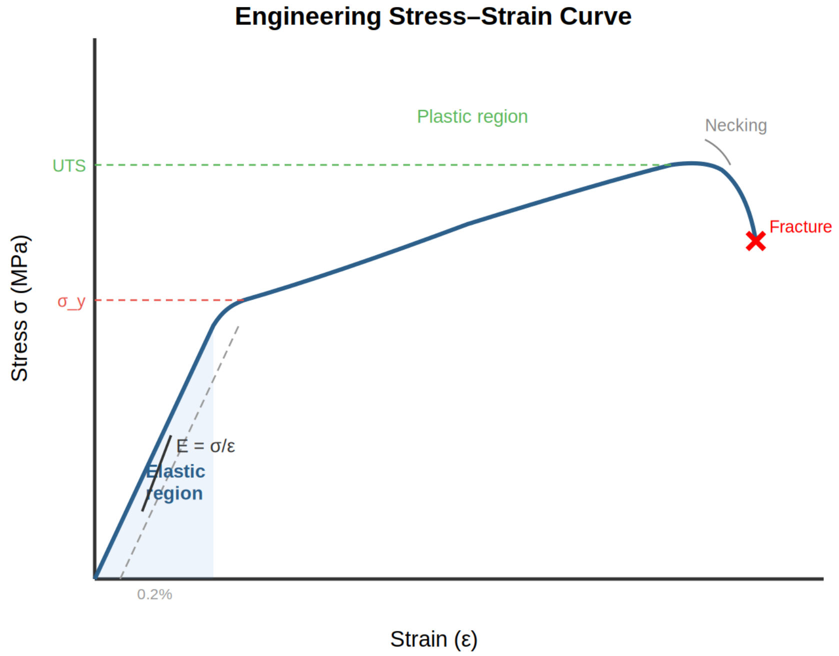
Normal stress (σ): Force perpendicular to the cross-section area. $\sigma = F / A_0$. Units: Pa (N/m²), MPa, GPa.

Shear stress (τ): Force parallel to the area of interest.

Engineering strain (ε): $\varepsilon = \Delta L / L_0 = (L_f - L_0) / L_0$ (dimensionless).

4.2 The Tensile Test

A unidirectional force is applied to a specimen (with defined gauge length L_0 and cross-section A_0) until fracture. Force and elongation are recorded and converted to stress and strain.



4.2.1 Young's Modulus (E)

The slope of the stress-strain curve in the elastic region (Hooke's Law):

$$\sigma = E \times \varepsilon \quad \rightarrow \quad E = \sigma / \varepsilon$$

Units: GPa. Elastic strain is reversible (no bond breakage). E depends directly on the bonding force — it is proportional to the slope of the interatomic force-separation curve at equilibrium spacing.

4.2.2 Yield Strength (σ_y)

The stress separating the elastic region (recoverable deformation) from the plastic region (permanent deformation).

0.2% offset method: Draw a line parallel to the elastic region starting at 0.2% strain (0.002). The intersection with the curve is σ_y. This standardizes the measurement.

Some steels (low-carbon): Show upper and lower yield points (discontinuous yielding).

4.2.3 Ultimate Tensile Strength (UTS)

Maximum stress the material can withstand before fracture. Beyond UTS, necking begins (localized thinning). Given in MPa.

Maximum force a specimen can sustain: $F_{max} = UTS \times A_0$

4.2.4 Ductility

Ability to be permanently deformed without breaking. Measured at fracture:

$$\% \text{ Elongation} = [(L_f - L_0) / L_0] \times 100$$

$$\% \text{ Reduction in Area} = [(A_0 - A_f) / A_0] \times 100$$

4.2.5 Poisson's Ratio (ν)

$$\nu = -\varepsilon_{lateral} / \varepsilon_{axial} \quad (\text{typically} \approx 0.3)$$

$$E = 2G(1 + \nu) \quad \text{where } G = \text{shear modulus}$$

4.2.6 True Stress and True Strain

$$\sigma_T = \sigma(1 + \varepsilon) \quad \text{and} \quad \varepsilon_T = \ln(1 + \varepsilon)$$

True stress uses the actual (instantaneous) cross-section, not the original. The true stress-true strain curve and engineering curve coincide up to the yield point but diverge after.

4.2.7 Toughness

The area under the true stress–true strain curve represents the energy per unit volume absorbed before fracture (tensile toughness). A material with lower yield strength but higher ductility can absorb more energy than a stronger but brittle material.

4.3 Impact Testing (Charpy Test)

Measures the energy absorbed by a notched specimen under sudden (high strain rate) loading.

$$K_{cv} = M \times g \times (h - h')$$

Where M = pendulum mass, g = gravity, h = initial height, h' = final height.

DBTT (Ductile-to-Brittle Transition Temperature): Temperature below which a material behaves in a brittle manner. BCC metals (like carbon steel) show a clear DBTT. FCC metals (like stainless steel) do NOT show a DBTT and remain tough at low temperatures. HCP metals have low absorbed energy throughout.

Notch sensitivity: Absorbed energy decreases as notch radius decreases (sharper notches = more brittle behavior).

Strain-rate sensitivity: Higher impact velocity → lower absorbed energy (more brittle).

4.4 Fatigue

Failure under dynamic, fluctuating stresses — can occur at stress levels well below σ_y or UTS. Brittle-like failure with no warning. Occurs by crack initiation and propagation. Fracture surfaces show beach marks/clamshell patterns.

4.4.1 Parameters Defining the Fatigue Cycle

Mean stress: $\sigma_m = (\sigma_{max} + \sigma_{min}) / 2$

Stress amplitude: $\sigma_a = (\sigma_{max} - \sigma_{min}) / 2$

Stress ratio: $R = \sigma_{min} / \sigma_{max}$

4.4.2 The S-N Curve

Fatigue/endurance limit: For some materials (steels), a stress below which the material will never fail by fatigue (horizontal asymptote in S-N curve).

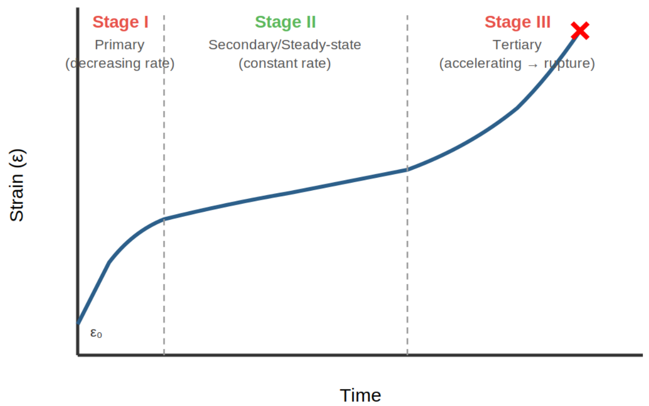
Fatigue life: Number of cycles at a given stress until failure.

Fatigue strength: Stress required to cause failure in a given number of cycles (e.g., 10^7).

Key exam application: Given a cantilever beam formula $\sigma = 32FL/(\pi d^3)$, calculate the stress, read cycles from S-N curve, and determine time to failure or required diameter for infinite life.

4.5 Creep

Time-dependent permanent deformation at elevated temperatures under constant stress (below σ_y).



Stage I (Primary): Decreasing creep rate (strain hardening dominates).

Stage II (Secondary/Steady-state): Constant creep rate (balance between hardening and recovery). The most important stage for design.

Stage III (Tertiary): Accelerating creep rate leading to rupture (necking, void formation).

Higher temperature and/or higher applied stress increase the creep rate and shorten rupture time.

4.6 Hardness Testing

Resistance to localized plastic deformation (indentation). A small indenter is forced into the surface under controlled load.

Test	Indenter	Load	Application
Brinell (HB)	10mm steel/WC ball	500–3000 kgf	Soft–medium metals
Rockwell (HR)	Diamond cone or ball	60–150 kgf	Metals, universal
Vickers (HV)	Diamond pyramid	1–120 kgf	All materials
Knoop (HK)	Diamond elongated	< 1 kgf	Micro-hardness, thin

UNIT 5: PLASTIC DEFORMATION AND STRENGTHENING

Plastic deformation in metals is due to dislocation movement. To strengthen (harden) a material, we must constrain or anchor dislocations. The trade-off: when hardening, ductility and toughness decrease. A compromise must be reached.

5.1 Grain Size Reduction Strengthening (Hall-Petch)

Dislocation motion is blocked at grain boundaries (slip systems of adjacent grains are not aligned). Smaller grains = more boundaries per unit volume = higher strength.

$$\sigma\_y = \sigma_0 + k\_y / \sqrt{d}$$

Where: σ_y = yield strength, σ_0 = intrinsic yield strength (constant for the material, like a monocrystal value), k_y = material constant related to dislocation formation, d = average grain size.

Key exam skill: Plot σ_y vs $1/\sqrt{d}$ to find σ_0 (intercept) and k_y (slope). Then predict σ_y for any grain size, or find d needed for a target σ_y.

5.2 Strain Hardening (Cold Work Strengthening)

Cold working is a forming operation where shape is changed by plastic deformation at temperatures below the recrystallization temperature. During deformation, a high number of new dislocations form and interact, making further movement more difficult.

5.2.1 Percent Cold Work (%CW)

$$\%CW = [(A_0 - A\_f) / A_0] \times 100$$

Where A_0 = original cross-sectional area, A_f = final area. For a rolling process where width is maintained:

$$\%CW = [(t_0 - t\_f) / t_0] \times 100 \quad \text{(thickness-based)}$$

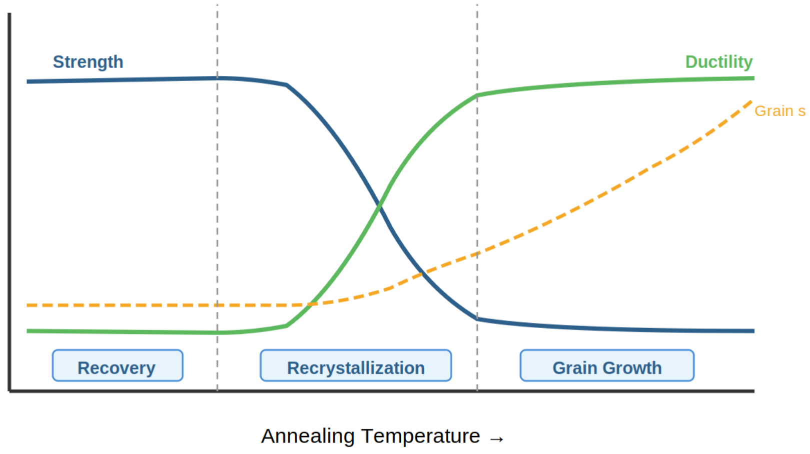
5.2.2 Effects of Cold Work

Increasing %CW produces: higher yield strength (σ_y), higher UTS, higher hardness, but lower ductility (% elongation), lower toughness, and decreased electrical conductivity and corrosion resistance. Grains stretch along deformation direction.

5.3 Annealing (Reversing Cold Work)

Heat treatment that allows mechanical properties to revert back to pre-cold-worked values. It does NOT change the shape of the specimen.

Annealing of Cold-Worked Metal: Three Stages



5.3.1 Stage 1: Recovery

• **No structural changes** (strength properties unchanged)

• **Dislocations reorganize** → internal stresses are relieved

• **Physical properties recovered** (electrical/thermal conductivity increases)

5.3.2 Stage 2: Recrystallization

New strain-free, equiaxed grains with low dislocation density form spontaneously. Mechanical properties are restored: ductility increases, strength decreases.

Recrystallization time (for 95% completion):

$$t_r = C_t \times (1/CW^n) \times \exp(Q_r / RT)$$

Where C_t = constant, CW = cold work fraction (e.g., 0.60 for 60%), n = exponent (often ~2), Q_r = activation energy, R = gas constant, T = temperature (K).

Recrystallization temperature: Temperature at which total recrystallization occurs in 1 hour. Typically 0.35–0.5 T_m (absolute). Higher CW lowers the recrystallization temperature and produces finer recrystallized grains. There exists a critical minimum CW (2–20%) below which recrystallization cannot occur. Alloying increases recrystallization temperature.

5.3.3 Stage 3: Grain Growth

If annealing continues beyond recrystallization, grains coarsen to reduce total grain boundary energy. Slight decrease in strength and slight increase in ductility. Growth rate increases exponentially with temperature and time.

5.3.4 Control of Grain Size

Fine grains: High %CW followed by recrystallization annealing (stop before grain growth).

Coarse grains: Minimum %CW followed by long recrystallization times (allow grain growth).

Grain size can be controlled by successive cold work + annealing cycles.

Key exam skill: Given two (t_r, T) pairs → find Q_r and C_t. Steps: 1) write two equations; 2) divide to eliminate C_t; 3) solve for Q_r; 4) substitute back to find C_t. Then predict t_r at any new T or CW.

5.4 Solid Solution Strengthening

Adding foreign atoms (alloying) that go into substitutional or interstitial solid solution makes dislocation movement more difficult.

• Greater size difference between lattice and alloying atom → greater strengthening.

• Increasing alloying element content → stronger alloy.

Example (Cu-Ni): Ni and Cu are fully soluble. Strength of pure Cu increases with Ni addition; strength of pure Ni increases with Cu addition. Maximum strength at ~60%Ni-40%Cu (Monel alloy).

Sn and Be in Cu: Much larger size difference from Cu → greater strengthening effect per atom compared to Ni or Zn.

KEY FORMULAS REFERENCE

Crystal Structure

$$BCC: a = 4R/\sqrt{3} \mid n = 2 \mid APF = 0.68$$
$$FCC: a = 2R\sqrt{2} \mid n = 4 \mid APF = 0.74$$
$$HCP: a = 2R \mid n = 6 \mid APF = 0.74 \mid c/a = 1.633$$
$$\text{Theoretical density: } \rho = nA / (V_c \times N_A)$$
$$\text{Interplanar spacing (cubic): } d(hkl) = a / \sqrt{(h^2+k^2+l^2)}$$
$$\text{Bragg's law: } n\lambda = 2d \sin\theta$$

Defects & Diffusion

$$\text{Vacancy concentration: } N_v/N = \exp(-Q_v/RT)$$
$$\text{Fick's 1st law: } J = -D(dC/dx)$$
$$\text{Diffusion coefficient: } D = D_0 \exp(-Q_d/RT)$$
$$\text{Fick's 2nd law: } (C_x-C_0)/(C_s-C_0) = 1 - \mathrm{erf}(x/(2\sqrt{(Dt)}))$$

Mechanical Properties

$$\text{Hooke's law: } \sigma = E\varepsilon$$
$$\text{Poisson: } \nu = -\varepsilon_{lat}/\varepsilon_{axial},\ E = 2G(1+\nu)$$
$$\text{True stress: } \sigma_T = \sigma(1+\varepsilon),\ \text{True strain: } \varepsilon_T = \ln(1+\varepsilon)$$
$$\%\ \text{Elongation} = (L_f-L_0)/L_0 \times 100$$
$$\%\ \text{Reduction in area} = (A_0-A_f)/A_0 \times 100$$
$$\text{Charpy: } K_{cv} = Mg(h-h')$$
$$\text{Cantilever beam: } \sigma = 32FL/(\pi d^3)$$

Strengthening

$$\text{Hall-Petch: } \sigma_y = \sigma_0 + k_y/\sqrt{d}$$
$$\%CW = (A_0-A_f)/A_0 \times 100$$
$$\text{Recrystallization time: } t_r = C_t(1/CW^n)\exp(Q_r/RT)$$

EXAM TIPS & FREQUENTLY TESTED TOPICS

Based on the practice problems provided, here are the most frequently tested concepts and the skills you need:

Unit 2 — Structure

1. Calculate theoretical density from unit cell parameters ($\rho = nA/V_cN_A$). Know n for BCC(2), FCC(4), HCP(6).

2. Convert between a and R for each structure. BCC: $a\sqrt{3}=4R$. FCC: $a\sqrt{2}=4R$.

3. Calculate linear and planar atomic densities for given directions/planes.

4. XRD problems: use Bragg's law + d(hkl) formula. Distinguish BCC vs FCC by d-spacing ratio of first two peaks (0.707 for BCC, 0.866 for FCC). Know diffraction rules.

5. From density and structure, calculate atomic molar mass (reverse density formula).

Unit 3 — Defects

1. Determine D_0 and Q_d from a log(D) vs 1/T plot (two points → slope and intercept).

2. Carburizing problems using Fick's 2nd law: find D from table, calculate erf argument, interpolate in erf table, solve for unknowns (t, T, x, or C_x).

3. Remember: $D \times t$ products can give you time at different temperatures.

Unit 4 — Mechanical Properties

1. Read E, σ_y, UTS, and ductility from stress-strain curves.

2. Calculate maximum load from UTS × A_0; check if deformation is elastic ($\sigma < \sigma_y$).

3. Material selection problems: check both σ_y (no plastic deformation) and E (stiffness/elongation limit).

4. Fatigue: calculate stress, read cycles from S-N curve, find time; for infinite life, use endurance limit.

Unit 5 — Strengthening

1. Hall-Petch: plot σ_y vs $1/\sqrt{d}$, extract constants, predict properties.

2. Calculate %CW for circular and rectangular cross-sections.

3. Read %CW charts to find acceptable range satisfying both strength and ductility requirements.

4. Design multi-step cold work + annealing processes when single-pass CW exceeds the material limit.

5. Recrystallization kinetics: two (t_r, T) data points → find Q_r and C_t; predict recrystallization at other conditions.

6. Identify recovery, recrystallization, and grain growth stages from property vs temperature data.